Domain randomization

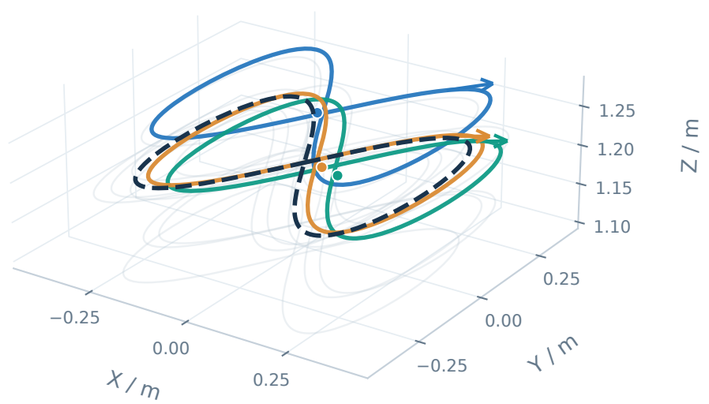
One figure-eight task, many starting conditions and drone responses

QUADROTOR + NMPC

12 sampled episodes | seeds 42–53

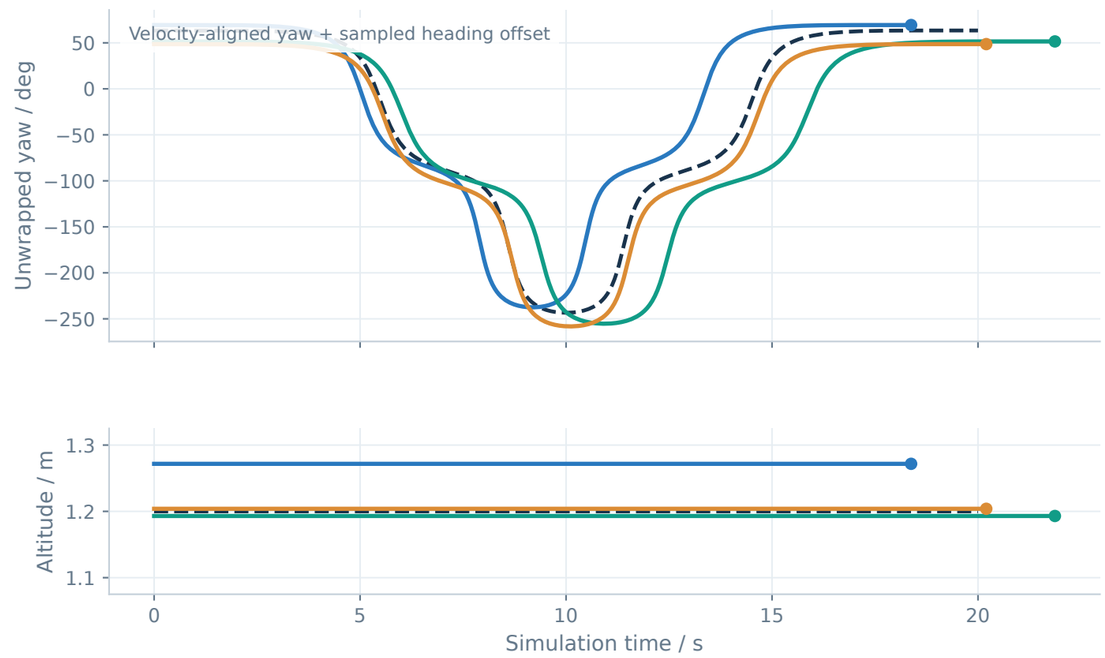
Nominal reference / model · Other sampled episodes · Seed 42 · Seed 47 · Seed 52

A Reference paths in Vicon coordinates

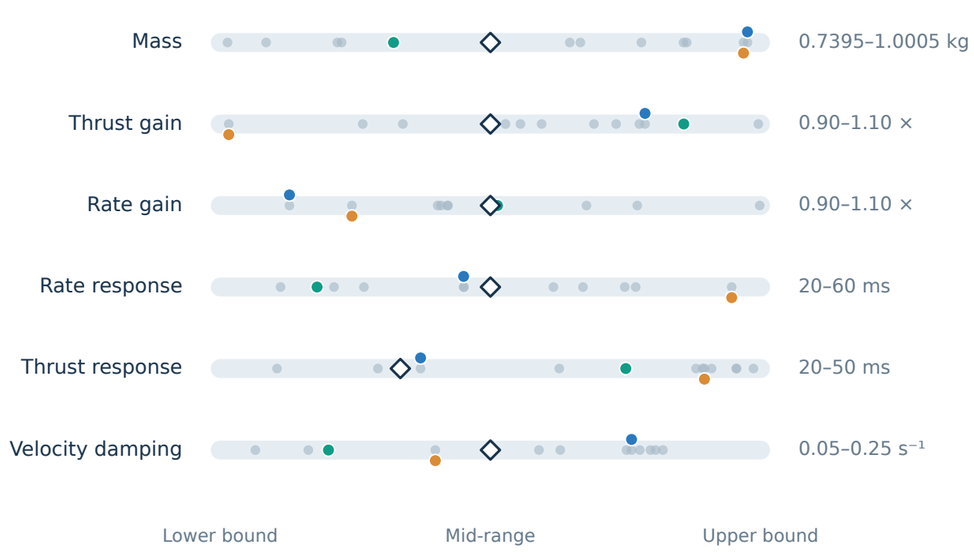


Arrows: desired heading at 23% of each cycle. Dots: perturbed initial positions.

B Heading, altitude and traversal time



C Sampled drone parameters



White diamonds: nominal MPC model. Colored dots: sampled actual plants.

SAMPLED ONCE PER EPISODE

MPC receives the reference and state; sampled physical parameters stay hidden. Appearance is unchanged.

D From sampled conditions to training pairs

01 **Sample an episode**
Task + initial errors + actual dynamics

▼

02 **Capture the current state**
Render RGB at the pre-action drone pose

▼

03 **Query the MPC expert**
Nominal prediction model → thrust + body rates

▼

04 **Save the synchronized pair**
RGB + state + reference → expert action label

Apply the action, advance the dynamics, then repeat steps 02–04.

Reference/settings visualization — no MPC rollout shown

Source: quadrotor_nmpc.json + randomization.py